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## EXPERIMENT 6

**Aim:** To build part of speech tagger using hmm model.

### Theory:

A Part-of-Speech (POS) tagger using the Hidden Markov Model (HMM) is a statistical approach to labeling each word in a sentence with its corresponding part-of-speech tag (e.g., noun, verb, adjective). HMMs are based on the concept of hidden states, where the true state (the POS tag) is unknown but can be inferred from observed data (the words). The model is trained on a labeled corpus, learning two key components: **transition probabilities**, which describe the likelihood of moving from one POS tag to another, and **emission probabilities**, which capture the likelihood of a word occurring given a specific POS tag. During tagging, the HMM computes the most probable sequence of tags for a sentence using algorithms like the Viterbi algorithm. This approach is effective because it leverages the sequential nature of language, where the tag of a word is often influenced by the tags of preceding words. Despite its simplicity, HMM-based POS tagging can perform well, especially when combined with additional features such as word context or capitalization.

### Code:

```
# Install nltk
!pip install nltk

# Import necessary modules
import nltk
from nltk.corpus import treebank
from nltk.tag import hmm
from nltk import FreqDist, ConditionalFreqDist
from nltk.tokenize import word_tokenize
from nltk import bigrams
```

Requirement already satisfied: nltk in /usr/local/lib/python3.11/dist-packages (3.9.1)  
Requirement already satisfied: click in /usr/local/lib/python3.11/dist-packages (from nltk) (8.1.8)  
Requirement already satisfied: joblib in /usr/local/lib/python3.11/dist-packages (from nltk) (1.4.2)  
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.11/dist-packages (from nltk) (2024.11.6)  
Requirement already satisfied: tqdm in /usr/local/lib/python3.11/dist-packages (from nltk) (4.67.1)

```
nltk.download('treebank')
nltk.download('universal_tagset')

# Get the tagged sentences from the Penn Treebank corpus
train_sents = treebank.tagged_sents()[:3000] # Use first 3000 sentences for training
test_sents = treebank.tagged_sents()[3000:]
```

[nltk\_data] Downloading package treebank to /root/nltk\_data...  
[nltk\_data] Unzipping corpora/treebank.zip.  
[nltk\_data] Downloading package universal\_tagset to /root/nltk\_data...  
[nltk\_data] Unzipping taggers/universal\_tagset.zip.



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```
[3] trainer = hmm.HiddenMarkovModelTrainer()  
    pos_tagger = trainer.train(train_sents)
```

```
[4] accuracy = pos_tagger.evaluate(test_sents)  
    print(f"Accuracy of the POS tagger: {accuracy:.4f}")
```

```
<ipython-input-4-740312787abd>:1: DeprecationWarning:  
  Function evaluate() has been deprecated. Use accuracy(gold)  
  instead.  
accuracy = pos_tagger.evaluate(test_sents)  
/usr/local/lib/python3.11/dist-packages/nltk/tag/hmm.py:333: RuntimeWarning: overflow encountered in cast  
X[i, j] = self._transitions[si].logprob(self._states[j])  
/usr/local/lib/python3.11/dist-packages/nltk/tag/hmm.py:335: RuntimeWarning: overflow encountered in cast  
O[i, k] = self._output_logprob(si, self._symbols[k])  
/usr/local/lib/python3.11/dist-packages/nltk/tag/hmm.py:331: RuntimeWarning: overflow encountered in cast  
P[i] = self._priors.logprob(si)  
/usr/local/lib/python3.11/dist-packages/nltk/tag/hmm.py:363: RuntimeWarning: overflow encountered in cast  
O[i, k] = self._output_logprob(si, self._symbols[k])  
Accuracy of the POS tagger: 0.3684
```

```
▶ sentence = "This is a sample sentence for POS tagging".split()  
  tagged_sentence = pos_tagger.tag(sentence)  
  print(tagged_sentence)
```

```
↻ [('This', 'DT'), ('is', 'VBZ'), ('a', 'DT'), ('sample', 'NNP'), ('sentence', 'NNP'), ('for', 'NNP'), ('POS', 'NNP'), ('tagging', 'NNP')]
```

## Conclusion:

Thus we successfully built part of speech tagger using hmm model.